AegisImage — Forensic Analysis Report

Version 0.1.0 | Generated: 2026-10-25T02:52:45.206213+00:00 | Schema: 1.0

EDUCATIONAL NOTICE: *AegisImage is an educational digital image forensics aid. Findings are exploratory observations and do not constitute proof of authenticity or tampering.*

Evidence Identity

Original Filename	sample_01_untouched.jpg	**Detected Format**	JPEG
File Size	8,050 bytes (7.9 KB)	**Dimensions**	240 x 180 px
Color Mode	RGB	**Frame Count**	1
SHA-256 Digest	`03ce18d6e71c53cf5504992759f6a3199cc66cae3ed296ad4c76b676ab7bebbd`		
SHA-1 Digest	`45ea55b63df4e61d35e867814742ab49f19003ae`	**MD5 Digest**	`04caaf2da37cf9b71223ff032a27e37c`

Metadata Inspection [Status: COMPLETED]

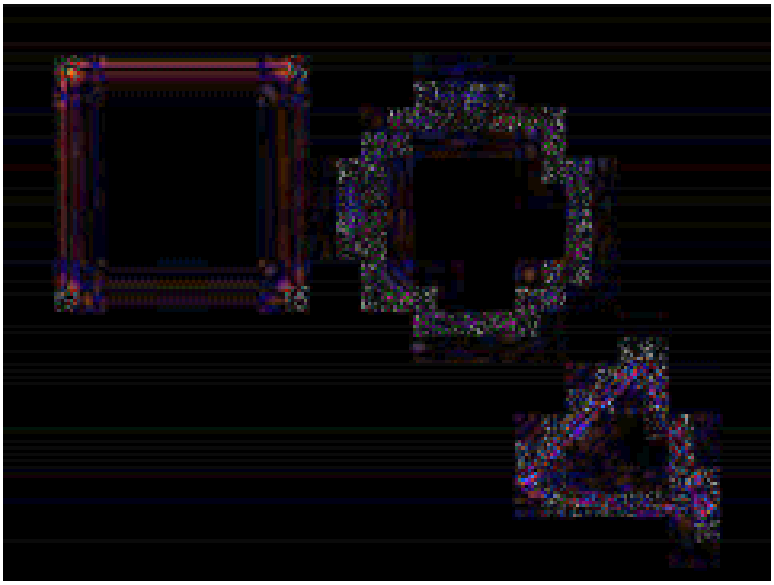
Observation: No EXIF metadata found. Many cameras, web services, and apps routinely strip metadata.

Limitations: Metadata may be absent, stripped, modified, or forged. The absence of metadata is not proof of tampering. | The absence of metadata is common across social media platforms and does not indicate tampering.

JPEG Error Level Analysis [Status: COMPLETED]

Observation: Error Level Analysis visualization generated with recompression quality 90 and amplification 10x.

Review Cue: ELA visualization highlights variations in compression error. Bright areas indicate higher differences between original and recompressed pixels; this naturally happens around contrast edges, fine textures, or spliced regions.

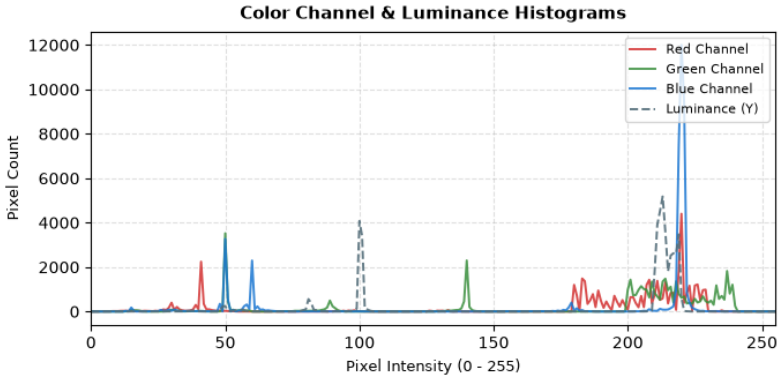


Derived Analysis Artifact — Not Original Evidence

Limitations: Error Level Analysis (ELA) is an exploratory visualization for JPEG images. Variations arise from compression history, high-frequency textures, and surface contrast. It does not provide a definitive tampering verdict. | ELA is an exploratory visualization and cannot provide a definitive authenticity or forgery verdict.

Image Statistics & Color Histograms [Status: COMPLETED]

Observation: Color channel distribution and descriptive statistics computed across 240x180 pixels (43,200 total).



Derived Analysis Artifact — Not Original Evidence

Limitations: Histograms describe pixel intensity distributions across color channels. | Histogram analysis is descriptive and cannot independently detect or prove image manipulation.

Standards & References

- *Krawetz, N. (2007). A Picture's Worth... Digital Image Analysis and Forensics. Black Hat DC.*
- *NIST Computer Forensic Tool Testing Program (CFTT).*
- *ISO/IEC 27037:2012 Guidelines for identification, collection, acquisition and preservation of digital evidence.*

Disclaimer: *This report is generated by AegisImage for educational and investigative guidance. It does not constitute legal proof of authenticity, chain-of-custody certification, or definitive tamper determination.*